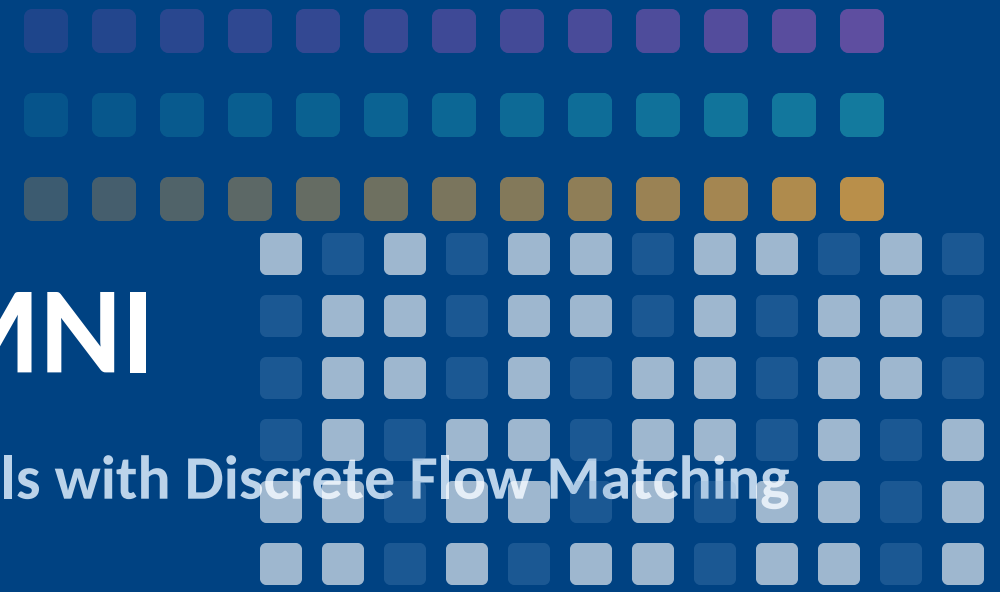




NExT-OMNI

Any-to-Any Omnimodal Foundation Models with Discrete Flow Matching

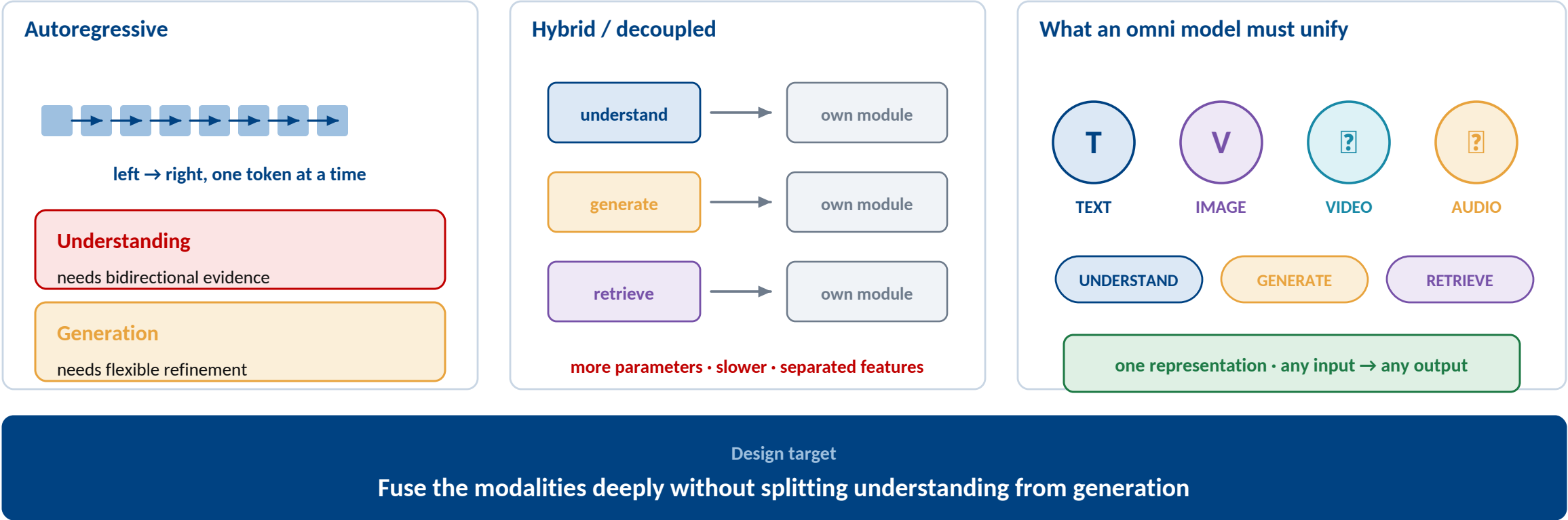
discrete flow • any-to-any

Run Luo • Xiaobo Xia • Lu Wang • Longze Chen • Renke Shan • Jing Luo • Min Yang • Tat-Seng Chua

SIAT, Chinese Academy of Sciences • UCAS • NExT++ Research Center • NUS

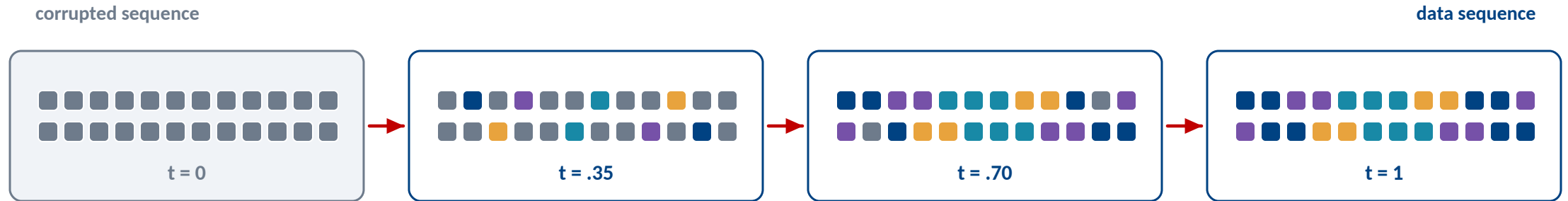
[arXiv:2510.13721v2 \[cs.CL\]](https://arxiv.org/abs/2510.13721v2)

The conflict — one model, three jobs, four modalities



Autoregression is sequential; hybrid systems unify tasks by **splitting the model**.

The alternative — discrete flow learns to correct in parallel



One bidirectional backbone predicts every clean token from the whole noisy sequence

$$x_t \sim p_t(\cdot | x_{-1}) \quad \text{model: } p_{1|t}(x_{-1} | x_t)$$

GLOBAL CORRUPTION

noise all positions

GLOBAL VELOCITY

move along the path

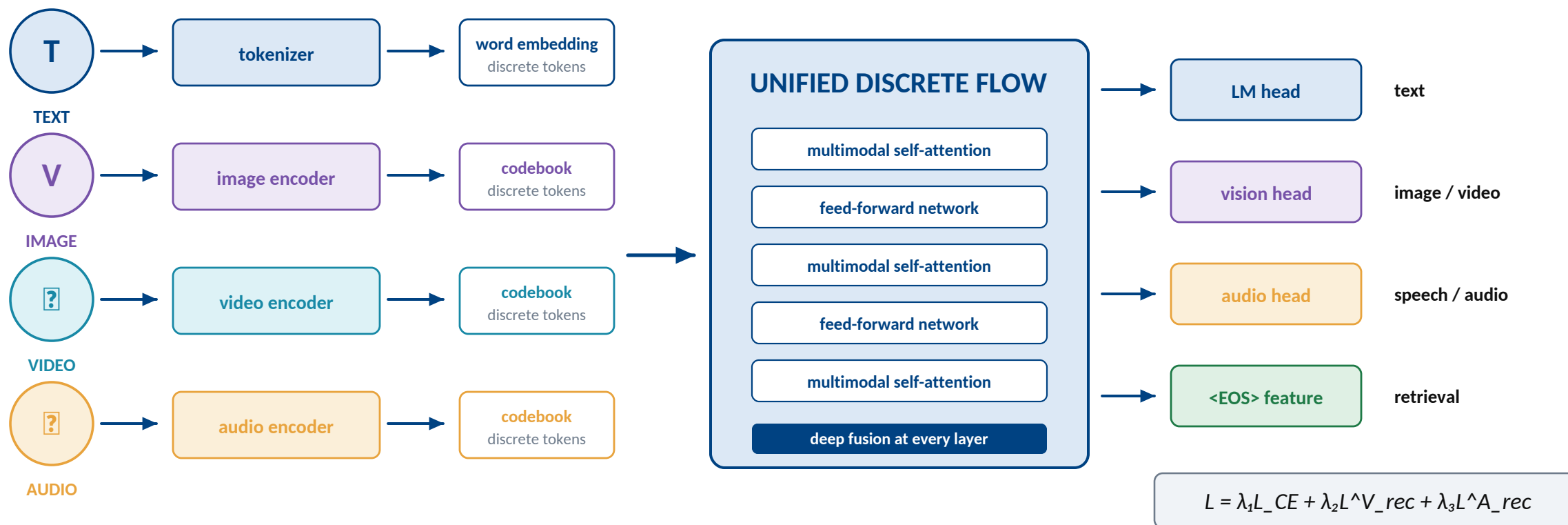
GLOBAL CORRECTION

refine in parallel

Unlike causal masking, every position can integrate context from both directions

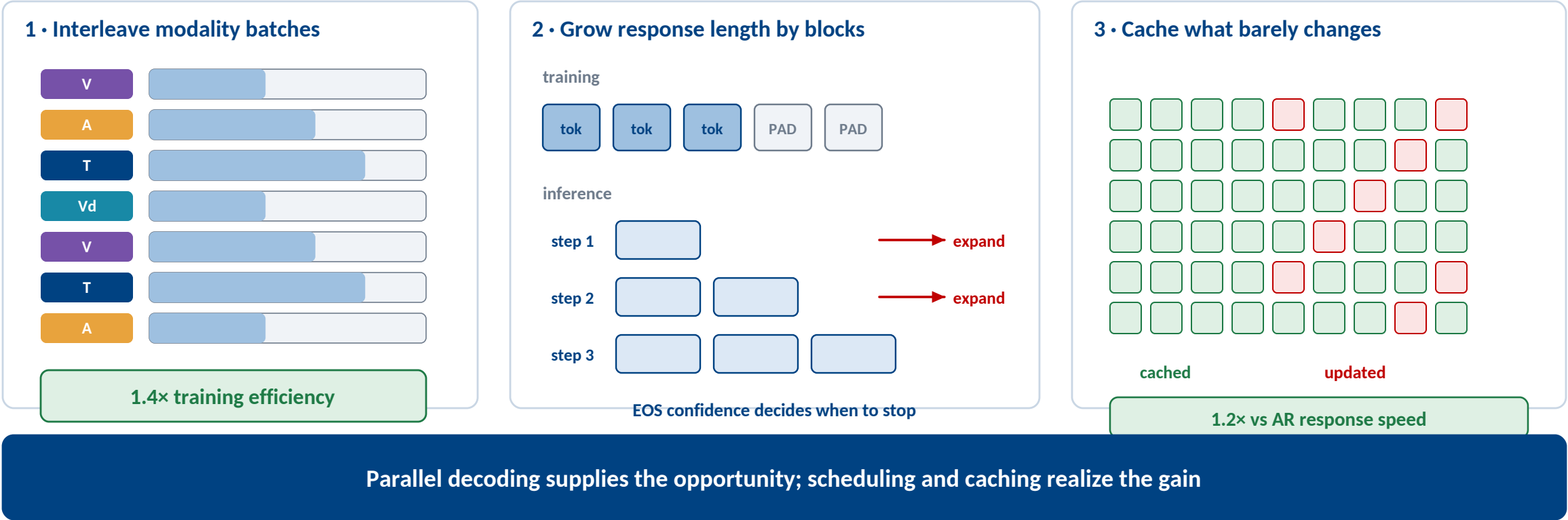
Corrupt globally, predict globally, refine iteratively — with **bidirectional context**.

NExT-OMNI — a single deeply fused representation



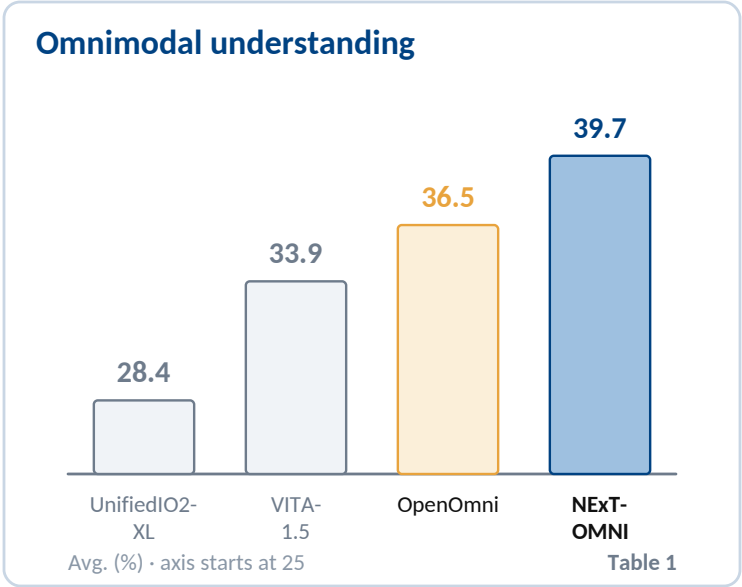
One backbone serves understanding, generation, and retrieval — **no task router required.**

Training and inference — make iterative decoding practical

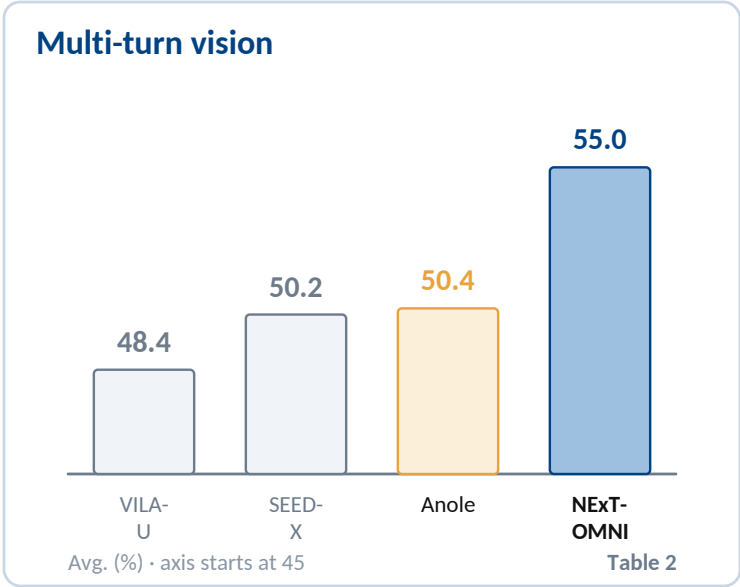


Interleave modalities, grow responses by blocks, and cache features that barely change.

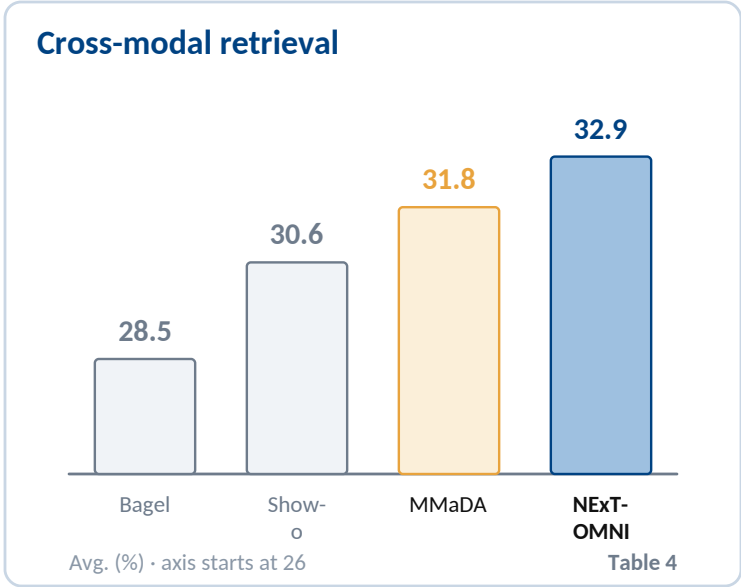
Evidence — the unified flow transfers across task families



+3.2 vs OpenOmni understanding



+4.6 vs next best multi-turn vision

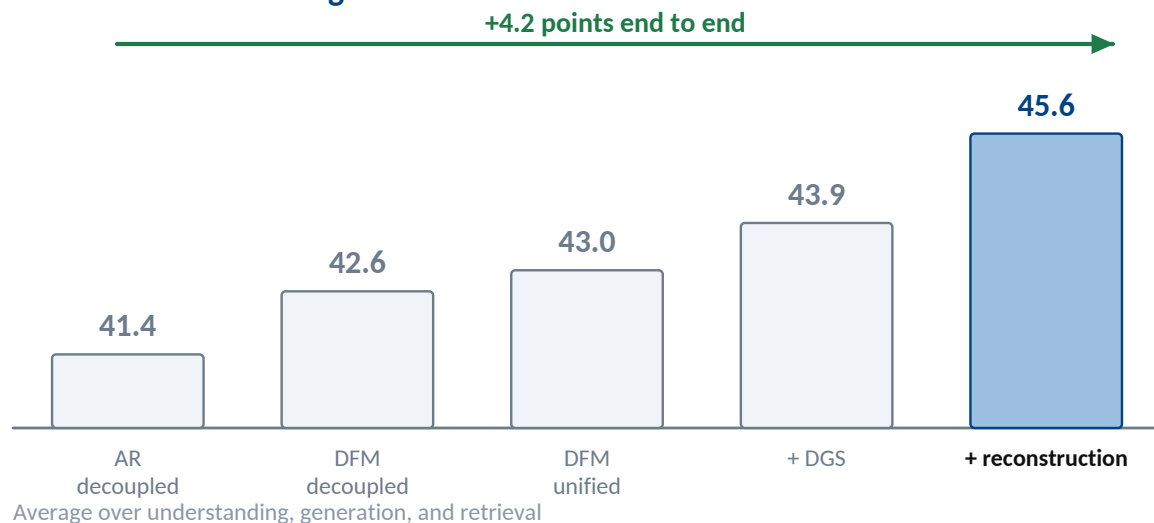


+1.1 vs MMaDA retrieval

NExT-OMNI leads on understanding, multi-turn vision, and cross-modal retrieval.

What the ablation says — unification needs the full recipe

Cumulative ablation average • Table 5



The recipe

- DFM** bidirectional correction
- Unified rep.** one feature space
- Dynamic length** restore understanding
- Reconstruction** retain fine detail

CONTRIBUTION

First open-source omni model fully built on DFM

CAPABILITY

Text • image • video • audio, any-to-any

NEXT STEP

Action trajectories and world-model video

DFM opens the door; dynamic length and reconstruction make the unified model **work.**